

Exam 1: Basics of R – Version A

Exam Agreement

This exam is an **individual assessment** and you should neither offer help to nor receive help from another person.

Cell phones are to be handed in to the instructor and all other electronic devices must be stored in bags at the front of the room.

No smart glasses or other wearable electronic devices are to be powered on during this exam except with accommodations.

You may use **no physical or electronic** resource to complete the work.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Exam Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

Computers must represent all information in binary, i.e. ones and zeros. Explain why this is necessary based on how computers are constructed.

Question 2: Command line and syntax

The following line of code does not work:

```
colors <- factor(my_colors, ordered = T levels = c("orange", "yellow", "blue"))
```

It produces the following error:

```
Error: unexpected symbol in "colors <- factor(my_colors, ordered=T levels"
```

Explain the source of the error. **Rewrite the line of code** that corrects the error and runs successfully below.

Question 3: Data Frames

Using the data frame `carnivora` (Fig. 1), what is the mean value of `FW` associated with the family `Canidae`?

carnivora		112 obs. of 17 variables
\$ Order	:	Factor w/ 1 level "Carnivora": 1 1 1 1 1 1...
\$ SuperFamily	:	Factor w/ 2 levels "Caniformia","Feliformi...
\$ Family	:	Factor w/ 8 levels "Ailuridae","Canidae",...
\$ Genus	:	Factor w/ 72 levels "Acinonyx","Acrtictis...
\$ Species	:	Factor w/ 112 levels "Acinonyx jubatus",...
\$ FW	:	num 31.1 9.7 10.6 7.2 22.2 13.8 2.9 3.9 1...
\$ SW	:	num 33.1 10.6 11.3 7.7 22 15.8 3.2 4.1 2...
\$ FB	:	num 130 84.5 53.5 52 128 95 37 43 24.5 33...
\$ SB	:	num 132.3 88.3 51.8 56.8 129 ...
\$ LS	:	num 5.5 6.2 4.3 3.8 8.8 4.3 7.1 4.8 3.5 4...
\$ GL	:	num 63 61.5 63.3 60 70.5 62 53.3 54.5 52 ...
\$ BW	:	num 425 225 NA NA 365 275 66 105 NA NA ...
\$ WA	:	int 35 98 NA 61 77 NA 21 56 NA NA ...
\$ AI	:	int NA NA NA 270 390 NA 165 225 NA NA ...
\$ LY	:	int 177 NA 127 NA 132 186 108 144 NA NA ...
\$ AM	:	Factor w/ 41 levels "", "1095", "1110", ...: 4...
\$ IB	:	Factor w/ 16 levels "", " S", "12", "13", ...: ...

Figure 1: The carnivora data set.

Question 4: Matrices and Arrays

You have a 4-dimensional array of dimensions 4, 5, 4, and 8 named `my_array`. How can you retrieve the first position of the first dimension, all of dimensions two and three, and the eighth position of the fourth dimension in a line of code?

Question 5: Lists

I have created a list stored with the name `my_list` with the following code:

```
my_list <- list(c("a", "b", "c", "d", "e"),  
               array(runif(300), dim = c(3, 10, 10)),  
               data.frame("x" = c(1.1, 1.2, 1.3),  
                           "y" = c(2.3, 5.6, 6.2)),  
               1L)
```

- a. How many members does this list contain?
- b. What data class is each member?
- c. How would you reference the letter “e” in `my_list`?

Question 6: Class Coercion

Write the function that is capable of making the necessary conversion:

- a. a character vector into a factor vector:
- b. a list to a vector:
- c. a numeric vector into a character vector:
- d. two vectors into a data frame:

Question 7: Data Classes and Special Values

You are working with the `lakers` data frame (Fig. 2), which is a data set of all the Lakers games played since 2008. Use code to complete the following questions:

lakers	34624 obs. of 13 variables
\$ date : int	20081028 20081028 20081028 20081028 20081028 2008102...
\$ opponent : chr	"POR" "POR" "POR" "POR" ...
\$ game_type: chr	"home" "home" "home" "home" ...
\$ time : chr	"12:00" "11:39" "11:37" "11:25" ...
\$ period : int	1 1 1 1 1 1 1 1 1 1 ...
\$ etype : chr	"jump ball" "shot" "rebound" "shot" ...
\$ team : chr	"OFF" "LAL" "LAL" "LAL" ...
\$ player : chr	" " "Pau Gasol" "Vladimir Radmanovic" "Derek Fisher" ...
\$ result : chr	" " "missed" " " "missed" ...
\$ points : int	0 0 0 0 0 2 0 1 0 2 ...

Figure 2: The lakers data set.

- You're concerned that the data set may be incomplete, but it's too large for you to scan. How can you check the entire data set to see if there are **any** missing values in it in one line of code that returns a single value?
- Assuming that there are missing values, how could you clean the data set so that only complete observations remain?

Question 8: Factors

Pick two columns from the `lakers` data frame (Fig. 2) and convert each to factored data, either ordered or unordered. The two columns' data must make sense as a factor. Replace the original column with the new factored column.

Question 9: Vectors

Create a vector of integers that counts from one to one thousand and multiple each item in this vector by the number 3 in one line of code.

Question 10: Troubleshooting

You need to use the function `http_credentials()` but you're unsure how. How can you access the help page for this function **with the console** (i.e., command line)?